# Compressor off-design report: summary and diagnostics

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Rotor-Rotor Same Shaft  |  run: Max_eta_Compressor_design_rr_same_shaft

Rotor-Rotor Same Shaft  |  PR_des = 1.5106, eta_des = 0.9586  |  correlations: polynomial, Howell factor 2.00

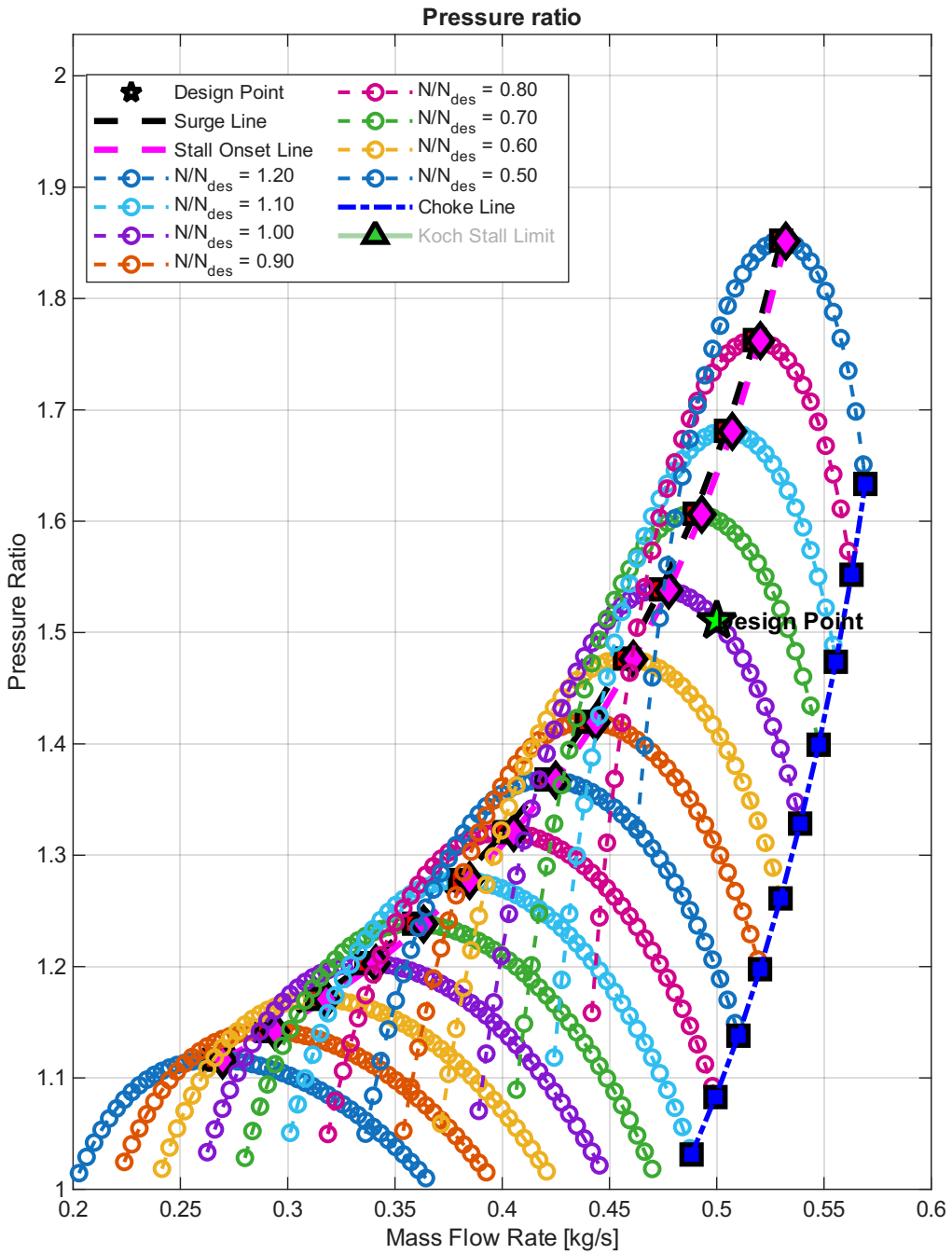
STABILITY (design speed): first limit = stall onset at m = 0.4776 kg/s (PR 1.538); stall margin SM = 6.6 %
   peak of the speed line (zero slope): m = 0.4740 kg/s | Howell stall onset: m = 0.4776 kg/s
   Koch stage limit: m = 0.5169 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5389 kg/s (Rotor 1 exit (relative)) -> choke margin 7.8 %, operating range (m_choke - m_lim)/m_choke = 11.4 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.378 -> surge if the installation has V_p/(A_c L_c) > 4.47 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (stall onset): highest loss rise Rotor 1 (omega/omega_des = 2.00), lowest de Haller Rotor 1 (0.78). Details: Tables >
      Blade rows.

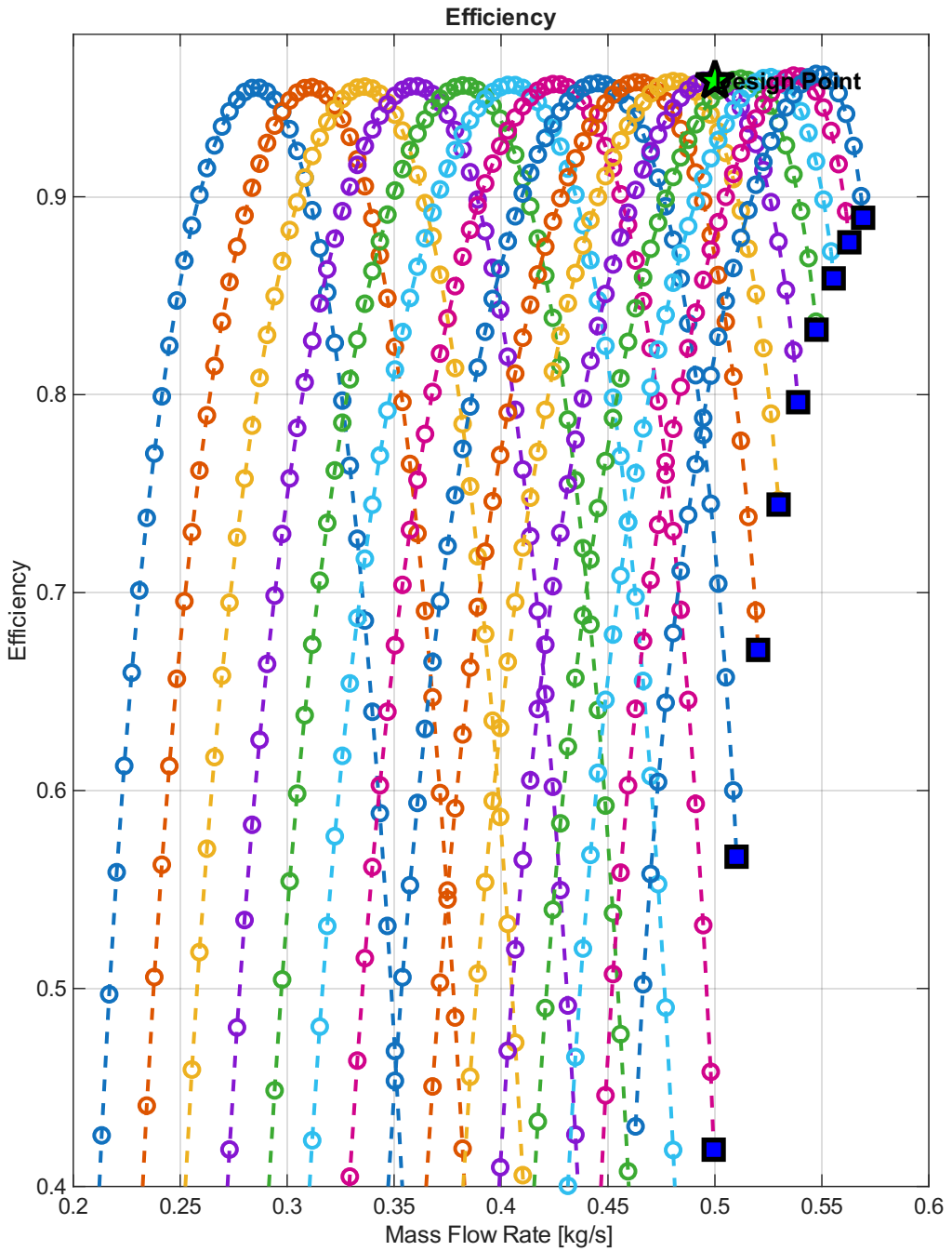
KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: Rotors R1+R2: L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in the same rotating frame (same shaft), so they
      are not re-energized; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.31)
     - RAISE the solidity x2.27 (stage L/g2 0.53 -> 1.20): R1 0.50 -> 1.13, R2 0.51 -> 1.16; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 35% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 1.21
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 1.23
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
       1 the ratio would be 0.91 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2), F_ef = [0.99 0.25] (data 0.39-1.11) -> indicative only
```

## Inputs of this run (.mat data and settings)

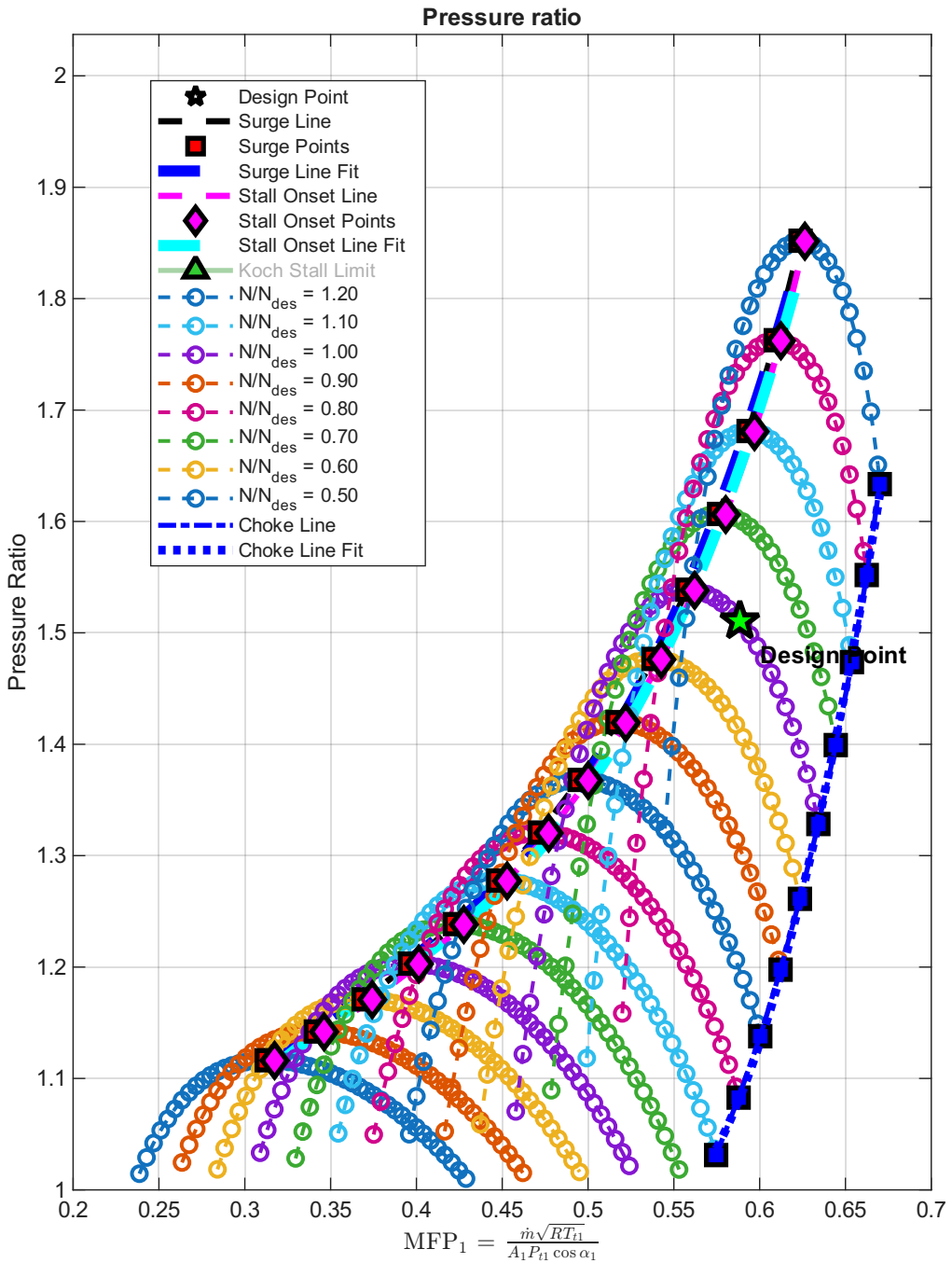
| Variable | Value | Unit | Source |
|---|---|---|---|
| Configuration | Rotor-Rotor same shaft | | |
| Input source | .mat: Max_eta_Compressor_design_rr_... | | |
| alpha1 | 30.3676 | deg | .mat: Max_eta_Compressor_design_rr_... |
| n_stages | 2 | - | .mat: Max_eta_Compressor_design_rr_... |
| A_vec | 0.00283128 0.00262365 0.00253207 | m^2 | .mat: Max_eta_Compressor_design_rr_... |
| beta1_dash | 50.3693 | deg | .mat: Max_eta_Compressor_design_rr_... |
| beta2_dash | 18.077 | deg | .mat: Max_eta_Compressor_design_rr_... |
| m_dot | 0.5 | kg/s | .mat: Max_eta_Compressor_design_rr_... |
| w_loss_rotor | 0.032576 | - | .mat: Max_eta_Compressor_design_rr_... |
| beta3_dash | 34.9249 | deg | .mat: Max_eta_Compressor_design_rr_... |
| beta4_dash | -0.0569908 | deg | .mat: Max_eta_Compressor_design_rr_... |
| deviation_1 | 12.2907 | deg | .mat: Max_eta_Compressor_design_rr_... |
| deviation_2 | 11.1084 | deg | .mat: Max_eta_Compressor_design_rr_... |
| incidence_1 | -6.04865 | deg | .mat: Max_eta_Compressor_design_rr_... |
| incidence_2 | -4.55729 | deg | .mat: Max_eta_Compressor_design_rr_... |
| deflection_1 | 13.953 | deg | .mat: Max_eta_Compressor_design_rr_... |
| deflection_2 | 19.3162 | deg | .mat: Max_eta_Compressor_design_rr_... |
| w_loss_rotor2 | 0.0293003 | - | .mat: Max_eta_Compressor_design_rr_... |
| U | 274.775 | m/s | .mat: Max_eta_Compressor_design_rr_... |
| sigma_1 | 0.500466 | - | .mat: Max_eta_Compressor_design_rr_... |
| sigma_2 | 0.512047 | - | .mat: Max_eta_Compressor_design_rr_... |
| chord | 0.01 | m | .mat: Max_eta_Compressor_design_rr_... |
| Correlation set | polynomial | | Run settings |
| Howell factor | 2 | - | Run settings |
| Design inlet Tt1 | 288 | K | Run settings |
| Design inlet pt1 | 100000 | Pa | Run settings |
| Speed lines N/N_des | 0.50 to 1.20 (15 lines) | - | Run settings |
| Mass-flow points per line | 129 | - | Run settings |
| Koch target Ch_ef/Ch_max | 0.85 | - | Run settings |
| B critical | 0.8 | - | Run settings |
| T_ref / p_ref (corrected map) | 288.15 / 101325 | K / Pa | Run settings |
| Koch tip clearance eps/g | 0.055 | - | Run settings |
| Koch axial gap dz/s | 0.38 | - | Run settings |

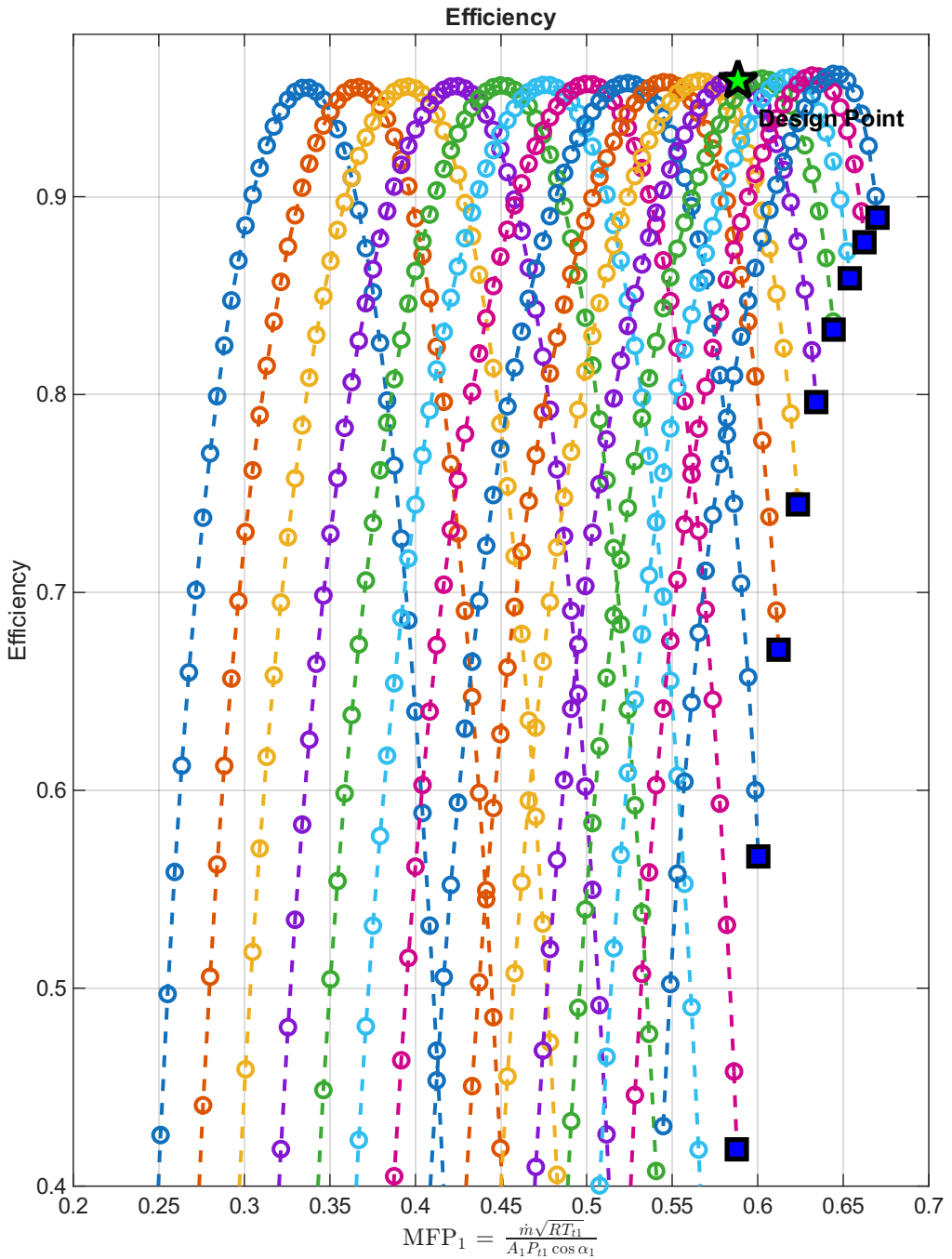
# Standard (dimensional) map - Rotor-Rotor Same Shaft



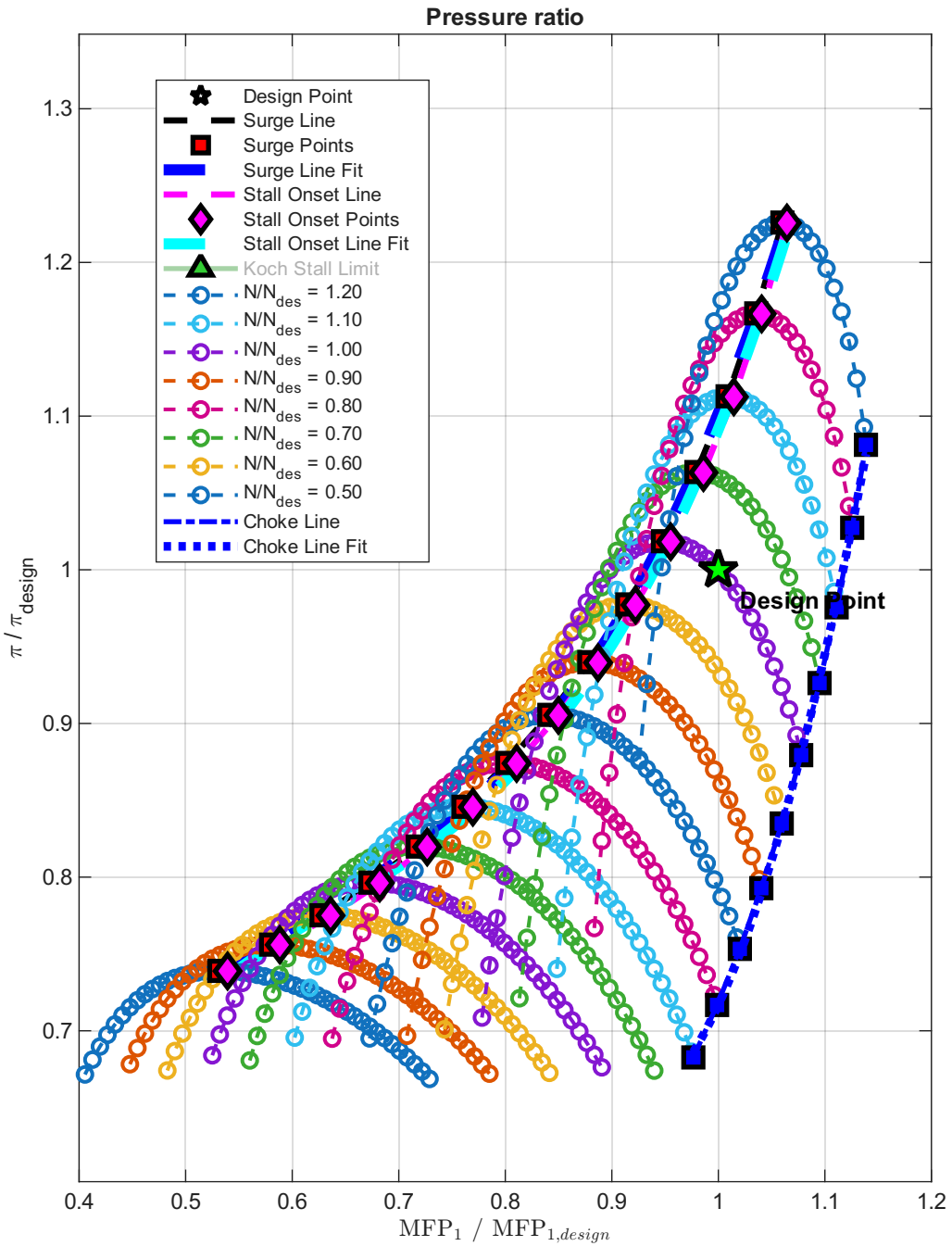



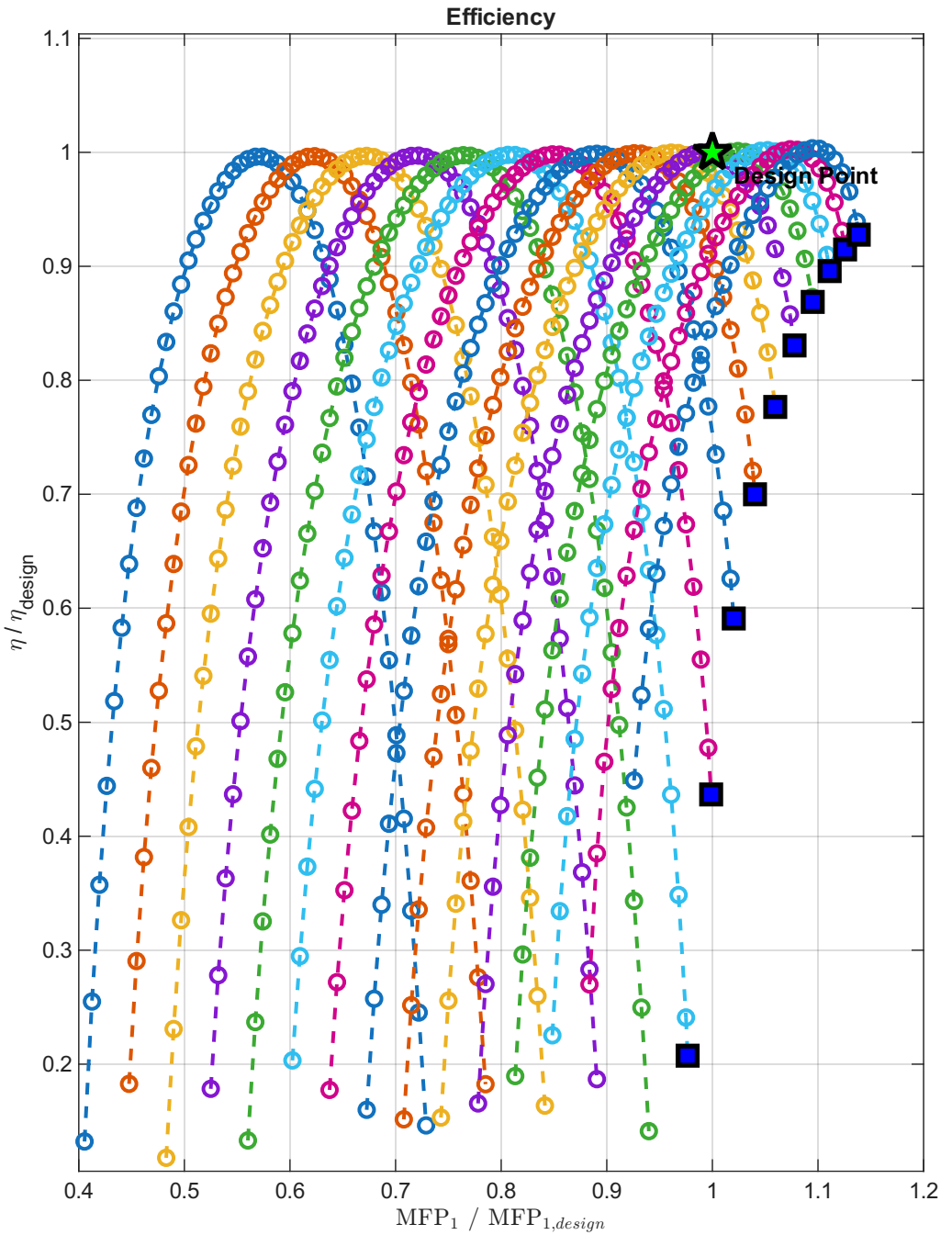
# Non-dimensional map - Rotor-Rotor Same Shaft



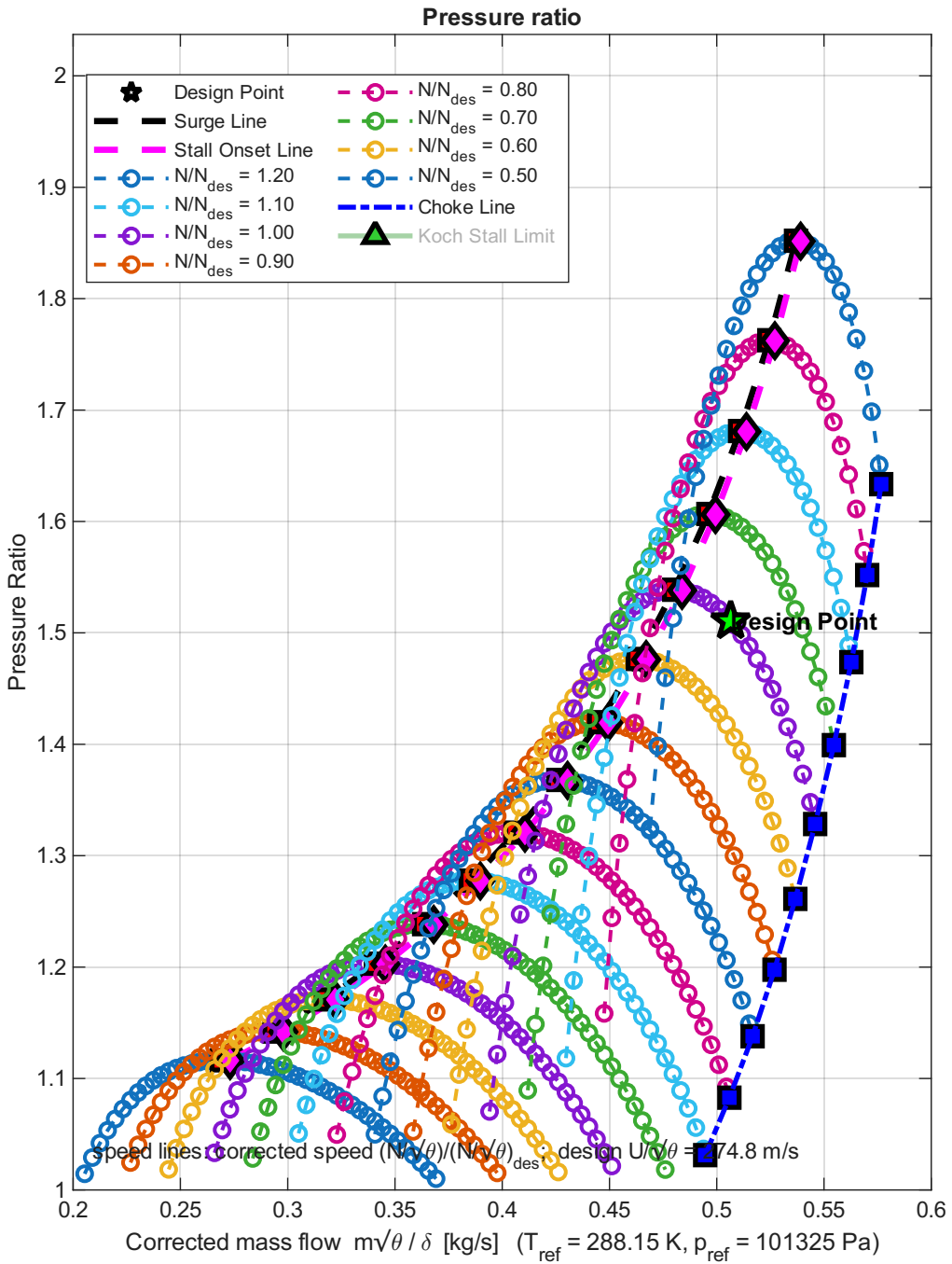



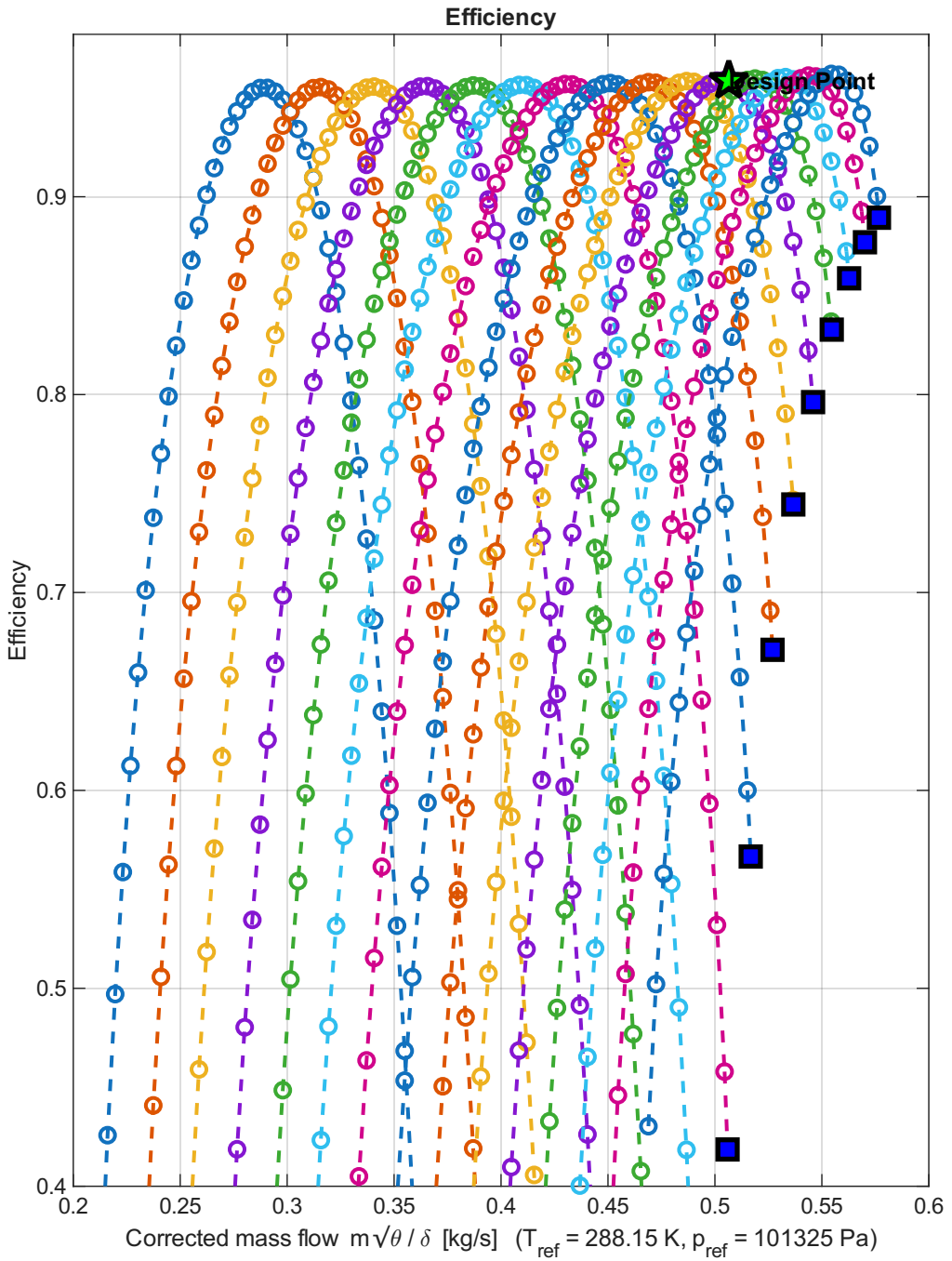
# Relative non-dimensional map - Rotor-Rotor Same Shaft



# Corrected map - Rotor-Rotor Same Shaft



## Fitted boundary equations (surge, stall onset, choke)

```
==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Same Shaft ====
Surge line:
Surge_PR = 22.7085*MFP^3 + -24.9854*MFP^2 + 10.2626*MFP + -0.3496   [R^2 = 0.99953]
Stall-onset line (Howell):
StallOnset_PR = 24.3762*MFP^3 + -27.3763*MFP^2 + 11.3359*MFP + -0.5092   [R^2 = 0.99945]
Choke line:
Choke_PR = 34.4661*MFP^2 + -36.7341*MFP + 10.7643   [R^2 = 0.99934]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Same Shaft ====
Surge line:
(PR/PR_design) = 3.0632*x^3 + -5.7274*x^2 + 3.9977*x + -0.2314   (x = MFP/MFP_design)   [R^2 = 0.99953]
Stall-onset line (Howell):
(PR/PR_design) = 3.2881*x^3 + -6.2754*x^2 + 4.4158*x + -0.3371   (x = MFP/MFP_design)   [R^2 = 0.99945]
Choke line:
(PR/PR_design) = 7.9006*x^2 + -14.3094*x + 7.1257   (x = MFP/MFP_design)   [R^2 = 0.99934]
```

## Design point

| Quantity | Value |
|---|---|
| Configuration | Rotor-Rotor Same Shaft |
| Pressure ratio PR_des | 1.5106 |
| Isentropic efficiency eta_des | 0.95864 |
| Mass flow m_des [kg/s] | 0.5 |
| MFP_1,des | 0.58846 |
| Corrected mass flow [kg/s] | 0.50649 |
| Inlet Tt1 [K] / Pt1 [Pa] | 288 / 100000 |
| Correlation set | polynomial |
| Howell factor | 2 |

## Choke

| N/N_des | m_choke [kg/s] | MFP_1 | PR | Eff | Choking row | Inside compressor region (PR>1) |
|---|---|---|---|---|---|---|
| 0.5 | 0.42954 | 0.5055 | 0.815 | | Rotor 1 exit (relative) | no |
| 0.55 | 0.44147 | 0.5196 | 0.852 | | Rotor 1 exit (relative) | no |
| 0.6 | 0.45347 | 0.5337 | 0.893 | | Rotor 1 exit (relative) | no |
| 0.65 | 0.46534 | 0.5477 | 0.936 | | Rotor 1 exit (relative) | no |
| 0.7 | 0.47696 | 0.5613 | 0.982 | | Rotor 1 exit (relative) | no |
| 0.75 | 0.48828 | 0.5747 | 1.031 | 0.199 | Rotor 1 exit (relative) | yes |
| 0.8 | 0.49942 | 0.5878 | 1.083 | 0.419 | Rotor 1 exit (relative) | yes |
| 0.85 | 0.51011 | 0.6004 | 1.138 | 0.567 | Rotor 1 exit (relative) | yes |
| 0.9 | 0.52018 | 0.6122 | 1.198 | 0.671 | Rotor 1 exit (relative) | yes |
| 0.95 | 0.52977 | 0.6235 | 1.261 | 0.744 | Rotor 1 exit (relative) | yes |
| 1 | 0.53889 | 0.6342 | 1.328 | 0.796 | Rotor 1 exit (relative) | yes |
| 1.05 | 0.54751 | 0.6444 | 1.399 | 0.833 | Rotor 1 exit (relative) | yes |
| 1.1 | 0.55553 | 0.6538 | 1.474 | 0.859 | Rotor 1 exit (relative) | yes |
| 1.15 | 0.56284 | 0.6624 | 1.552 | 0.877 | Rotor 1 exit (relative) | yes |
| 1.2 | 0.56929 | 0.67 | 1.633 | 0.889 | Rotor 1 exit (relative) | yes |

## Stall onset (Howell)

| N/N_des | m_stall [kg/s] | PR | Eff | Stalling row |
|---|---|---|---|---|
| 0.5 | 0.26966 | 1.116 | 0.936 | Rotor 1 |
| 0.55 | 0.29417 | 1.142 | 0.936 | Rotor 1 |
| 0.6 | 0.31799 | 1.171 | 0.936 | Rotor 1 |
| 0.65 | 0.34107 | 1.203 | 0.937 | Rotor 1 |
| 0.7 | 0.36336 | 1.238 | 0.937 | Rotor 1 |
| 0.75 | 0.3848 | 1.277 | 0.937 | Rotor 1 |
| 0.8 | 0.40534 | 1.32 | 0.938 | Rotor 1 |
| 0.85 | 0.42494 | 1.368 | 0.938 | Rotor 1 |
| 0.9 | 0.44355 | 1.419 | 0.939 | Rotor 1 |
| 0.95 | 0.46113 | 1.476 | 0.939 | Rotor 1 |
| 1 | 0.47763 | 1.538 | 0.94 | Rotor 1 |
| 1.05 | 0.49302 | 1.606 | 0.941 | Rotor 1 |
| 1.1 | 0.50727 | 1.681 | 0.941 | Rotor 1 |
| 1.15 | 0.52035 | 1.762 | 0.942 | Rotor 1 |
| 1.2 | 0.53222 | 1.852 | 0.943 | Rotor 1 |

## Peak (zero slope)

| N/N_des | m_peak [kg/s] | PR | Eff | m_stall - m_peak [kg/s] | First limit on throttling |
|---|---|---|---|---|---|
| 0.5 | 0.26579 | 1.116 | 0.925 | 0.00386 | stall onset (Howell) |
| 0.55 | 0.2901 | 1.142 | 0.926 | 0.00407 | stall onset (Howell) |
| 0.6 | 0.31376 | 1.171 | 0.926 | 0.00423 | stall onset (Howell) |
| 0.65 | 0.33674 | 1.203 | 0.927 | 0.00434 | stall onset (Howell) |
| 0.7 | 0.35897 | 1.239 | 0.927 | 0.0044 | stall onset (Howell) |
| 0.75 | 0.3804 | 1.278 | 0.928 | 0.0044 | stall onset (Howell) |
| 0.8 | 0.401 | 1.321 | 0.929 | 0.00435 | stall onset (Howell) |
| 0.85 | 0.4207 | 1.368 | 0.93 | 0.00424 | stall onset (Howell) |
| 0.9 | 0.43947 | 1.42 | 0.931 | 0.00408 | stall onset (Howell) |
| 0.95 | 0.45726 | 1.477 | 0.932 | 0.00386 | stall onset (Howell) |
| 1 | 0.47404 | 1.539 | 0.933 | 0.00359 | stall onset (Howell) |
| 1.05 | 0.48974 | 1.607 | 0.934 | 0.00328 | stall onset (Howell) |
| 1.1 | 0.50435 | 1.681 | 0.935 | 0.00292 | stall onset (Howell) |
| 1.15 | 0.51781 | 1.763 | 0.937 | 0.00254 | stall onset (Howell) |
| 1.2 | 0.53008 | 1.852 | 0.938 | 0.00213 | stall onset (Howell) |

## Koch limit

| N/N_des | m_Koch [kg/s] | PR | Eff | Limiting stage |
|---|---|---|---|---|
| 0.5 | 0.28353 | 1.112 | 0.955 | Rotors R1+R2 |
| 0.55 | 0.31076 | 1.137 | 0.955 | Rotors R1+R2 |
| 0.6 | 0.33748 | 1.163 | 0.955 | Rotors R1+R2 |
| 0.65 | 0.36356 | 1.192 | 0.955 | Rotors R1+R2 |
| 0.7 | 0.38891 | 1.223 | 0.954 | Rotors R1+R2 |
| 0.75 | 0.41338 | 1.257 | 0.952 | Rotors R1+R2 |
| 0.8 | 0.43679 | 1.293 | 0.95 | Rotors R1+R2 |
| 0.85 | 0.459 | 1.331 | 0.947 | Rotors R1+R2 |
| 0.9 | 0.47988 | 1.371 | 0.943 | Rotors R1+R2 |
| 0.95 | 0.49923 | 1.414 | 0.939 | Rotors R1+R2 |
| 1 | 0.51694 | 1.459 | 0.933 | Rotors R1+R2 |
| 1.05 | 0.53286 | 1.505 | 0.927 | Rotors R1+R2 |
| 1.1 | 0.5468 | 1.553 | 0.92 | Rotors R1+R2 |
| 1.15 | 0.55861 | 1.603 | 0.912 | Rotors R1+R2 |
| 1.2 | 0.56808 | 1.655 | 0.903 | Rotors R1+R2 |

## Stability limit & Greitzer B

| N/N_des | First limit | m_lim [kg/s] | PR | U [m/s] | a_p [m/s] | U/(2a_p) | (V_p/(A_c L_c))_crit | B | Beyond the limit |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | stall onset | 0.26966 | 1.116 | 137.4 | 345.9 | 0.1986 | 16.23 | NaN | - |
| 0.55 | stall onset | 0.29417 | 1.142 | 151.1 | 347.1 | 0.2177 | 13.51 | NaN | - |
| 0.6 | stall onset | 0.31799 | 1.171 | 164.9 | 348.4 | 0.2366 | 11.44 | NaN | - |
| 0.65 | stall onset | 0.34107 | 1.203 | 178.6 | 349.9 | 0.2552 | 9.82 | NaN | - |
| 0.7 | stall onset | 0.36336 | 1.238 | 192.3 | 351.4 | 0.2737 | 8.55 | NaN | - |
| 0.75 | stall onset | 0.3848 | 1.277 | 206.1 | 353.1 | 0.2918 | 7.51 | NaN | - |
| 0.8 | stall onset | 0.40534 | 1.32 | 219.8 | 354.8 | 0.3097 | 6.67 | NaN | - |
| 0.85 | stall onset | 0.42494 | 1.368 | 233.6 | 356.7 | 0.3274 | 5.97 | NaN | - |
| 0.9 | stall onset | 0.44355 | 1.419 | 247.3 | 358.7 | 0.3447 | 5.39 | NaN | - |
| 0.95 | stall onset | 0.46113 | 1.476 | 261 | 360.9 | 0.3617 | 4.89 | NaN | - |
| 1 | stall onset | 0.47763 | 1.538 | 274.8 | 363.1 | 0.3784 | 4.47 | NaN | - |
| 1.05 | stall onset | 0.49302 | 1.606 | 288.5 | 365.5 | 0.3947 | 4.11 | NaN | - |
| 1.1 | stall onset | 0.50727 | 1.681 | 302.3 | 367.9 | 0.4107 | 3.79 | NaN | - |
| 1.15 | stall onset | 0.52035 | 1.762 | 316 | 370.5 | 0.4264 | 3.52 | NaN | - |
| 1.2 | stall onset | 0.53222 | 1.852 | 329.7 | 373.3 | 0.4417 | 3.28 | NaN | - |

# Blade rows

| Operating point (N = 1) | m [kg/s] | Blade row | Incidence i [deg] | (i - i_des)/deflection | Loss omega | omega/omega_des | M_in (row frame) | M_out (row frame) | de Haller V_out/V_in | Deviation [deg] | Watch |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Design point | 0.5 | Rotor 1 | -6.05 | 0 | 0.0326 | 1 | 0.75 | 0.611 | 0.829 | 12.29 | |
| Design point | 0.5 | Rotor 2 | -4.56 | 0 | 0.0293 | 1 | 0.611 | 0.533 | 0.879 | 11.11 | |
| First limit (stall onset) | 0.47763 | Rotor 1 | -2.85 | 0.229 | 0.0652 | 2 | 0.736 | 0.566 | 0.784 | 13.07 | Howell stall |
| First limit (stall onset) | 0.47763 | Rotor 2 | -3.78 | 0.04 | 0.0302 | 1.032 | 0.566 | 0.492 | 0.876 | 11.19 | |
| Near choke (0.99 m_choke) | 0.5335 | Rotor 1 | -11.51 | -0.391 | 0.1242 | 3.813 | 0.782 | 0.821 | 1.045 | 12.29 | |
| Near choke (0.99 m_choke) | 0.5335 | Rotor 2 | -4.55 | 0 | 0.0293 | 1 | 0.821 | 0.669 | 0.831 | 11.11 | |